Southern New Hampshire University

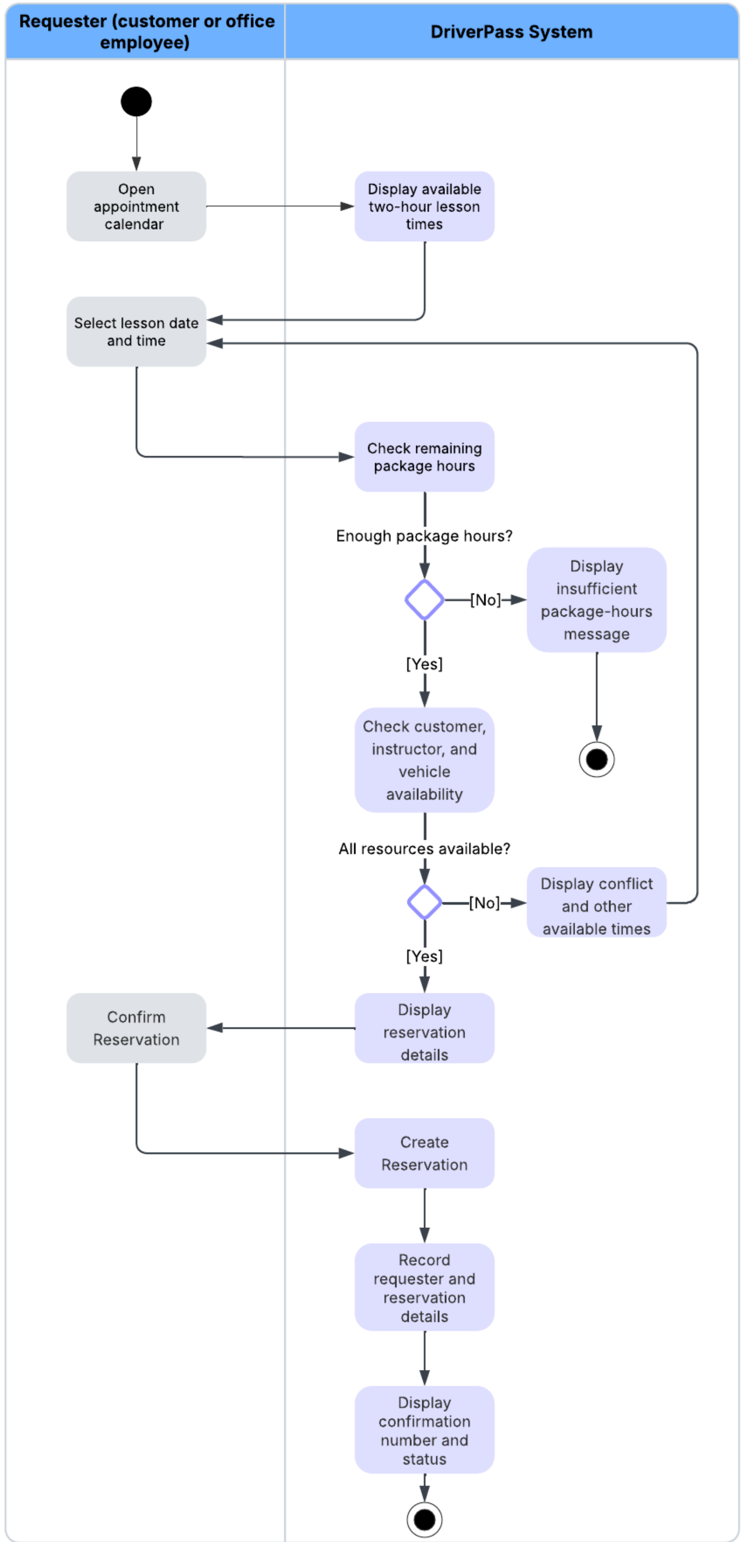
DriverPass System Design Document

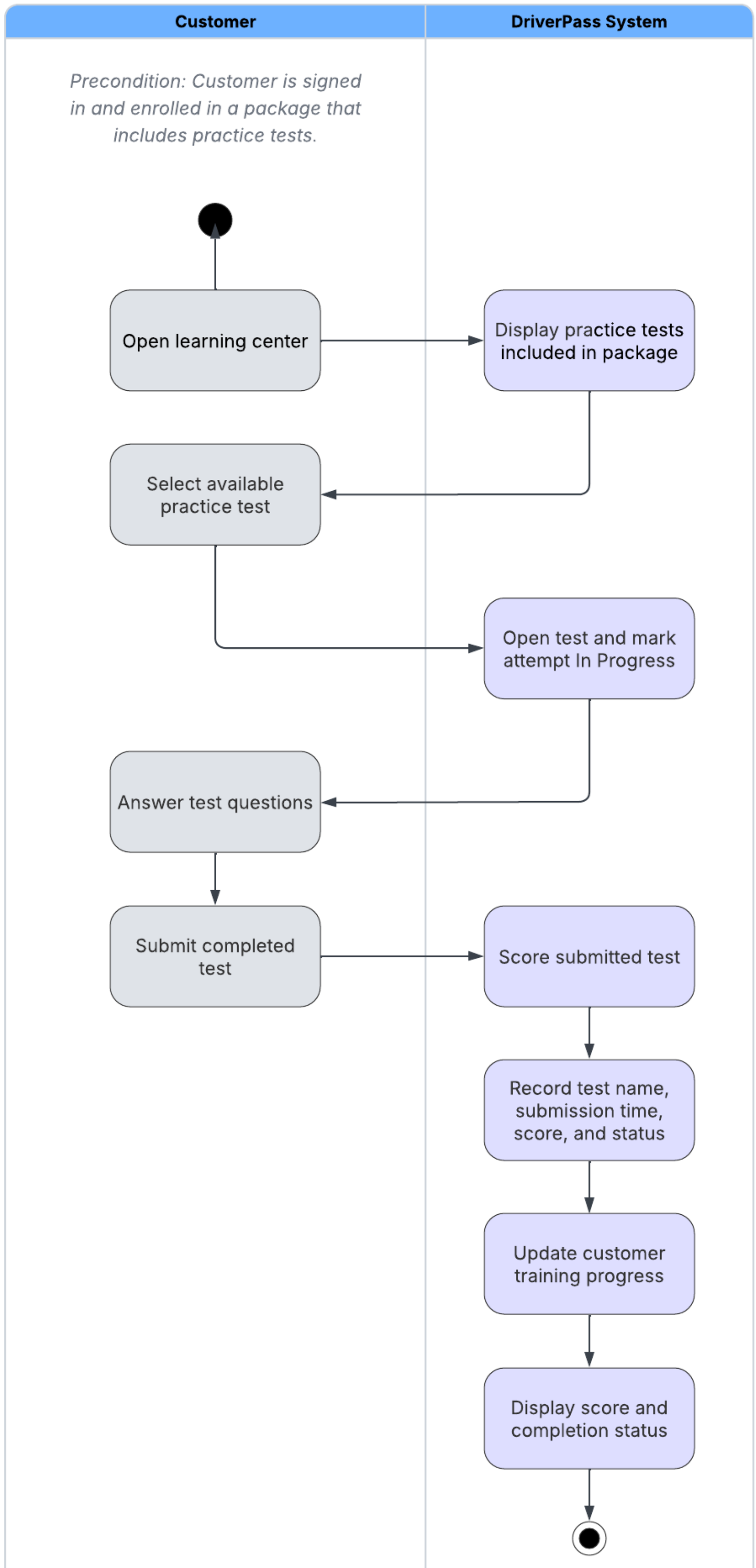
UML Diagrams

UML Use Case Diagram

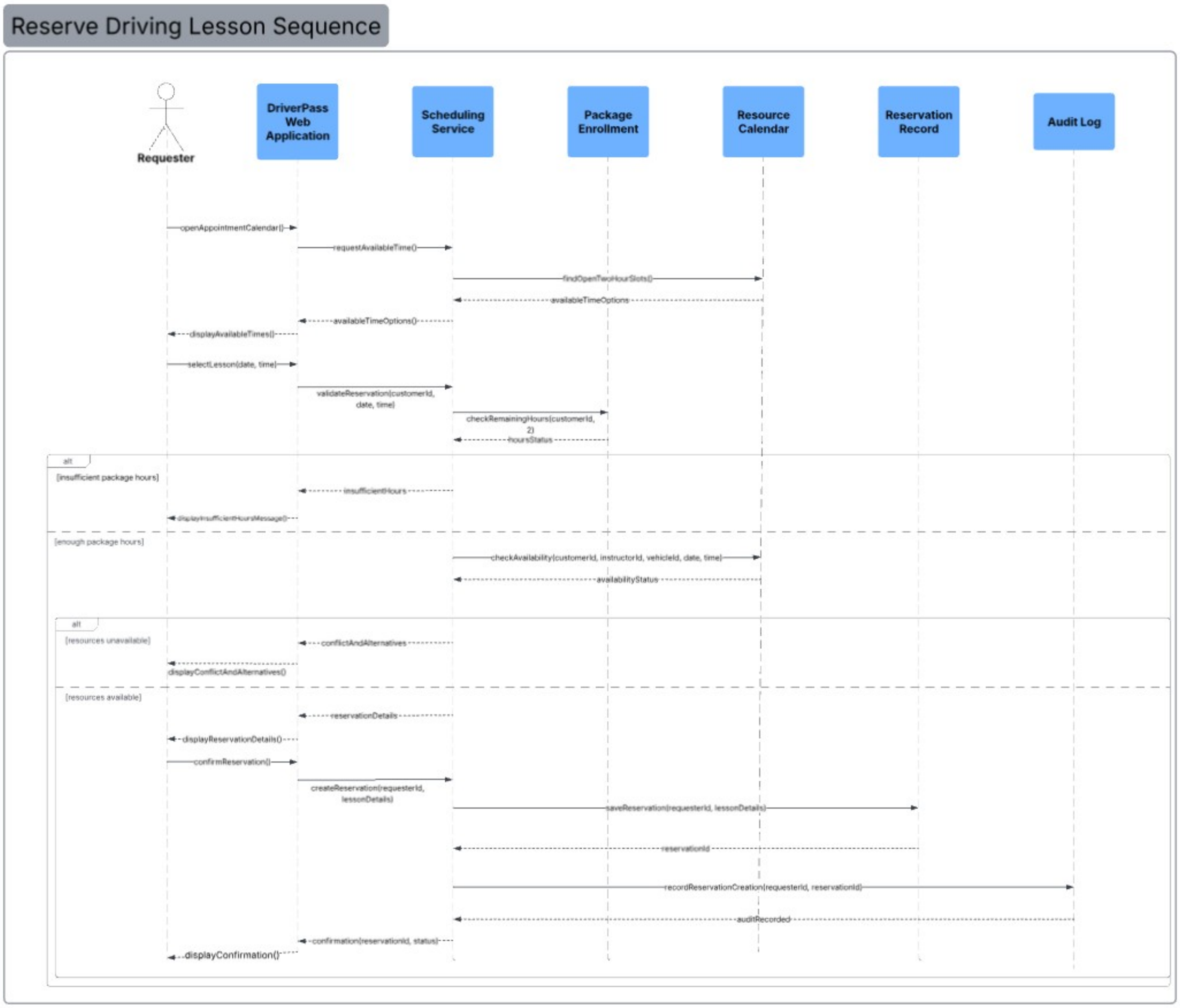

DriverPass Web Application
Email Provider
Payment Provider
Customer
Owner
Office Employee
Authorized Content Staff
Driving Instructor
IT Personnel
Recover Forgotten Password
Purchase Training Package
Review Available Training Packages
Register Customer account
Update Account Information
Access Online Training Materials
Complete Practice Tests
Review Training Progress
View Available Lesson Times
Reserve Driving Lesson
Sign In to System
Cancel or Reschedule Lesson
Maintain Customer Records
VIew Assigned Lessons & Customer Information
Document Lesson Completion and Feedback
View Customer and Business Data
Generate and Export Reports
Activate or Deactivate Training Packages
Review and Publish Training Content
Manage User Accounts
Configure Roles and Access Permissions
Review System Activity Logs

UML Activity Diagrams

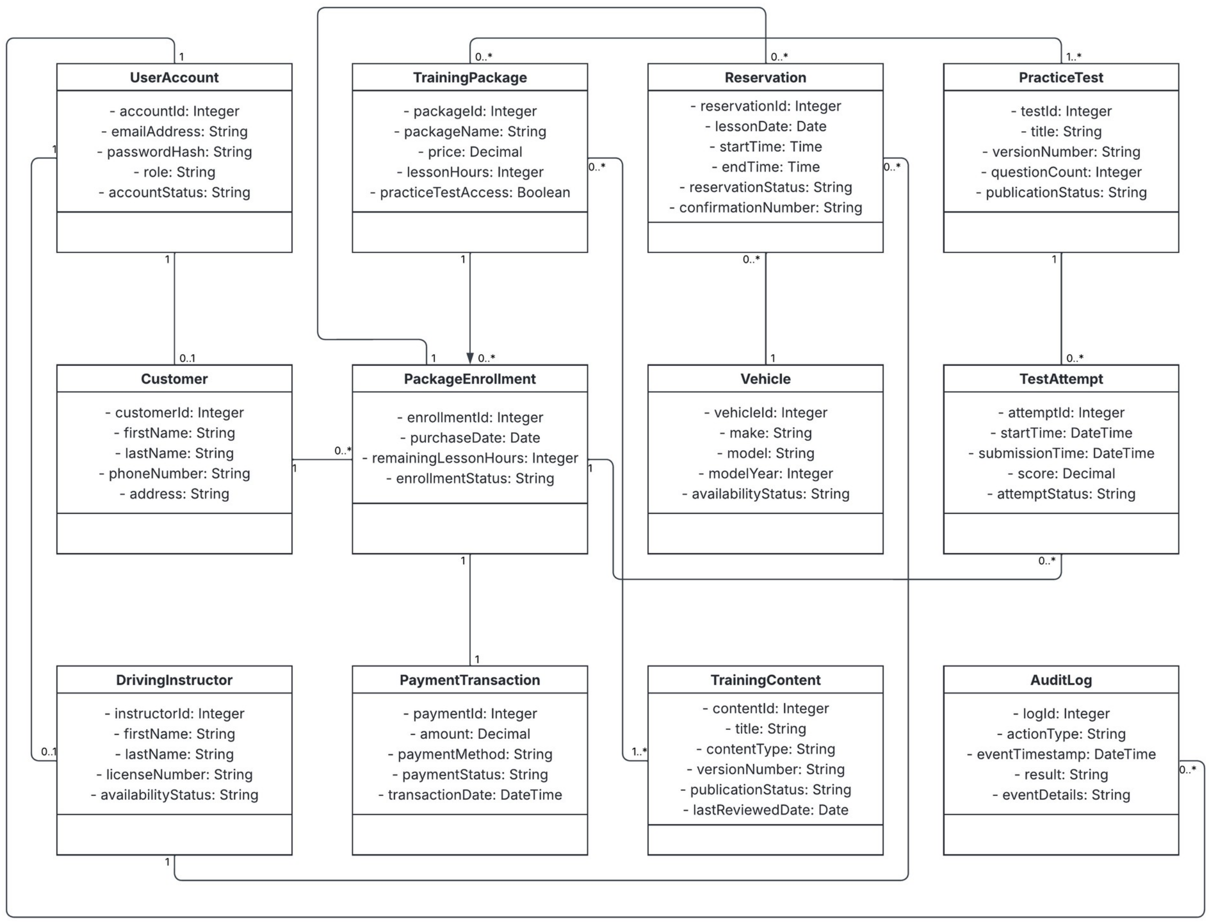



Customer
DriverPass System
Precondition: Customer is signed in and enrolled in a package that includes practice tests.
Open learning center
Display practice tests included in package
Select available practice test
Open test and mark attempt In Progress
Answer test questions
Submit completed test
Score submitted test
Record test name, submission time, score, and status
Update customer training progress
Display score and completion status

UML Sequence Diagram



UML Class Diagram



Technical Requirements

Hardware

DriverPass will use scalable cloud-based computing resources rather than relying on a single physical server at the business location. The hosting environment will need enough processing power, memory, and storage to run the web application, store customer and business records, and support backups. Exact server capacity should be determined after DriverPass provides expected user and transaction volumes. Customers and employees will access the system through internet-connected desktop computers, laptops, tablets, or smartphones.

Software

The system will be a browser-based application, so customers and employees will not need to install a native mobile or desktop application. It will support the two most recent versions of Chrome, Edge, Firefox, and Safari. The application will need software for account management, package purchases, online training, practice tests, lesson scheduling, reporting, and role-based access. A relational database will store user accounts, customer records, package enrollments, payments, reservations, test results, training content, and security logs. Secure connections will be required for the outside payment and email providers.

Development and Support Tools

The development team will need a programming environment, source-control repository, UML modeling software, automated testing tools, and a controlled process for building and releasing updates. Testing should cover account access, payments, scheduling conflicts, package-hour validation, browser compatibility, and user permissions. Monitoring and logging tools will help the support team identify failed sign-ins, application errors, performance problems, and unusual account activity. Authorized content staff will also need tools to review and publish updated training materials because the first release will not receive automatic updates from the DMV.

Infrastructure

The application and database should be hosted in a secure cloud environment with encrypted network connections, regular backups, system monitoring, and the ability to increase capacity as usage grows. The design should target 99.5% monthly availability, excluding scheduled maintenance. Pages and account information should normally load within four seconds, while scheduling and confirmation actions should complete within eight seconds. HTTPS, salted nonreversible password storage, role-based permissions, inactivity controls, single-use password recovery links, and security logs will protect customer and business information. The infrastructure must also support reliable connections to the payment and email providers without exposing sensitive payment or account information.